Overview of Federated learning

Subject: Federated learning

1.

Heterogeneous federated learning Most of the existing federated learning strategies assume that local models share the same global model architecture. Recently, a new federated learning framework named HeteroFL was developed to address heterogeneous clients equipped with very different computation and communication capabilities. The HeteroFL technique can enable the training of heterogeneous local models with dynamically varying computation and non-IID data complexities while still producing a single accurate global inference model.

2.

f (x 1 , … , x K) = 1 K ∑ i = 1 K f i (x i) {\displaystyle f(\mathbf {x} _{1},\dots ,\mathbf {x} _{K})={\dfrac {1}{K}}\sum _{i=1}^{K}f_{i}(\mathbf {x} _{i})}

3.

Use cases Federated learning typically applies when individual actors need to train models on larger datasets than their own, but cannot afford to share the data in itself with others (e.g., for legal, strategic or economic reasons). The technology yet requires good connections between local servers and minimum computational power for each node.

4.

Personalized federated learning by pruning (Sub-FedAvg) Federated learning methods have poor global performance under non-IID settings. This motivates clients to yield personalized models in federation. To change this, Vahidian et al. recently introduced the algorithm Sub-FedAvg which does hybrid pruning (structured and unstructured pruning) with averaging on the intersection of clients' drawn subnetworks. This simultaneously addresses communication efficiency, resource constraints and personalized models accuracies. Sub-FedAvg also extends the "lottery ticket hypothesis" of centrally trained neural networks to federated learning, with the research question: "Do winning tickets exist for clients' neural networks being trained in federated learning? If so, how to effectively draw the personalized subnetworks for each client?" Sub-FedAvg experimentally answers "yes" and proposes two algorithms to effectively draw the personalized subnetworks.

5.

Centralized federated learning In the centralized federated learning setting, a central server is used to orchestrate the different steps of the algorithms and coordinate all the participating nodes during the learning process. The server is responsible for the nodes selection at the beginning of the training process and for the aggregation of the received model updates. Since all the selected nodes have to send updates to a single entity, the server may become a bottleneck of the system.

6.

where K {\displaystyle K} is the number of nodes, x i {\displaystyle \mathbf {x} _{i}} are the weights of model as viewed by node i {\displaystyle i} , and f i {\displaystyle f_{i}} is node i {\displaystyle i} 's local objective function, which describes how model weights x i {\displaystyle \mathbf {x} _{i}} conforms to node i {\displaystyle i} 's local dataset. The goal of federated learning is to train a common model on all of the nodes' local datasets, in other words:

7.

Those parameters have to be optimized depending on the constraints of the machine learning application (e.g., available computing power, available memory, bandwidth). For instance, stochastically choosing a limited fraction C {\displaystyle C} of nodes for each iteration diminishes computing cost and may prevent overfitting, in the same way that stochastic gradient descent can reduce overfitting.

8.

Personalized federated learning In many application domains, not all clients are the same. Clients may share the same model architecture, but each have a different data distribution—which is known as Personalized federated learning (PFL), or even have different model architectures, like mobile phones and IoT devices—which is known as Heterogeneous federated learning. The key challenge in training a PFL model is to keep each client different from others, and at the same time benefit from generalizing across clients.

9.

Industry 4.0: smart manufacturing In Industry 4.0, there is a widespread adoption of machine learning techniques to improve the efficiency and effectiveness of industrial process while guaranteeing a high level of safety. Nevertheless, privacy of sensitive data for industries and manufacturing companies is of paramount importance. Federated learning algorithms can be applied to these problems as they do not disclose any sensitive data. In addition, FL also implemented for PM2.5 prediction to support Smart city sensing applications.

10.

Medicine: digital health Federated learning seeks to address the problem of data governance and privacy by training algorithms collaboratively without exchanging the data itself. Today's standard approach of centralizing data from multiple centers comes at the cost of critical concerns regarding patient privacy and data protection. To solve this problem, the ability to train machine learning models at scale across multiple medical institutions without moving the data is a critical technology. Nature Digital Medicine published the paper "The Future of Digital Health with Federated Learning" in September 2020, in which the authors explore how federated learning may provide a solution for the future of digital health, and highlight the challenges and considerations that need to be addressed. Recently, a collaboration of 20 different institutions around the world validated the utility of training AI models using federated learning. In a paper published in Nature Medicine "Federated learning for predicting clinical outcomes in patients with COVID-19", they showcased the accuracy and generalizability of a federated AI model for the prediction of oxygen needs in patients with COVID-19 infections. Furthermore, in a published paper "A Systematic Review of Federated Learning in the Healthcare Area: From the Perspective of Data Properties and Applications", the authors trying to provide a set of challenges on FL challenges on medical data-centric perspective. A coalition from industry and academia has developed MedPerf, an open source platform that enables validation of medical AI models in real world data. The platform relies technically on federated evaluation of AI models aiming to alleviate concerns of patient privacy and conceptually on diverse benchmark committees to build the specifications of neutral clinically impactful benchmarks.

11.

Non-IID data In most cases, the assumption of independent and identically distributed samples across local nodes does not hold for federated learning setups. Under this setting, the performances of the training process may vary significantly according to the unbalanced local data samples as well as the particular probability distribution of the training examples (i.e., features and labels) stored at the local nodes. To further investigate the effects of non-IID data, the following description considers the main categories presented in the preprint by Peter Kairouz et al. from 2019. The description of non-IID data relies on the analysis of the joint probability between features and labels for each node. This allows decoupling of each contribution according to the specific distribution available at the local nodes. The main categories for non-iid data can be summarized as follows: